

# How to find $A^{-1}$

## Why elementary matrices?

Every elementary matrix is invertible and if  $E$  is elementary then  $EA$  is the same as performing the elementary row operation which got  $E$  on  $A$ .

## An algorithm for finding $A^{-1}$

- 1 Start with the matrix  $[A \ I]$ .
- 2 Row reduce to reduced row echelon form.
- 3 If the reduced row echelon form is  $[I \ B]$  then  $B$  is  $A^{-1}$ .
- 4 Otherwise  $A$  is not invertible.

Elementary matrices.

- ① Exchange rows in the identity matrix

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \leftarrow \text{elem.}$$

$\uparrow$   
E

EA For my is just  
exchanging rows 2 and 3  
in A.

- ② Multiply by a non-zero scalar.

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 1 \end{pmatrix} \leftarrow \text{multiplies the third row by 4}$$

③ Adding a mult. of one row to another.

$$\left( \begin{array}{cccc} 1 & 0 & 3 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{array} \right) \leftarrow \text{add } 3 \times R_3 + R_1.$$

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Ex: (the inverse algorithm).

$$\left( \begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & 3 & 0 & 1 & 0 \\ 1 & 3 & 6 & 0 & 0 & 1 \end{array} \right)$$

$\begin{matrix} \uparrow & & \uparrow \\ A & & I_3 \end{matrix}$

ref  $\hookleftarrow$

$$\left( \begin{array}{ccc|ccc} 1 & 0 & 0 & 0 & -3 & 1 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 & -1 \end{array} \right)$$

$\uparrow$   
ref of  
 $A$ .

$A$  is not  
invertible  
because the  
ref( $A$ ) is  
not  $I_3$ .

Why does this algorithm work?

If we row reduce  $A$  to  $I$  then there is a series of elem. matrices  $E_1, \dots, E_n$  s.t.

$$E_n \cdots E_3 E_2 E_1 A = I.$$

$$E_n^{-1} E_n \cdots E_1 A = E_n^{-1} I$$

$$E_n \cdots E_1 A = E_n^{-1}$$

$$\text{so } A = \underline{E_1^{-1} E_2^{-1} \cdots E_n^{-1}}$$

$$\text{so } A^{-1} = E_n E_{n-1} \cdots E_2 E_1$$

← product of invert. matrices  
and so is invertible.

# The Invertible Matrix Theorem

## Theorem (The many equivalences of being invertible)

*If  $A$  is an  $n \times n$  matrix then the following are equivalent:*

- (a)  $A$  is invertible.
- (b)  $A$  can be row reduced to the identity matrix.
- (c)  $A$  has  $n$  pivot positions.
- (d)  $Ax = 0$  has only the trivial solution.
- (e) The columns of  $A$  form a linearly independent set.
- (f) The linear transformation  $x \mapsto Ax$  is one-to-one.
- (g) The columns of  $A$  span  $\mathbb{R}^n$ .
- (h) The linear transformation  $x \mapsto Ax$  is onto.
- (i) There is an  $n \times n$  matrix  $C$  such that  $CA = I$ .
- (j) There is an  $n \times n$  matrix  $D$  such that  $AD = I$ .
- (k)  $A^T$  is invertible.

! } !

Claim:  $A$  is invertible iff  $A^T$  is invertible.

Pf/ Suppose  $B$  is such that  $AB = BA = I$ .

$$\text{So } (AB)^T = (BA)^T = I.$$

$$\text{and then we get } B^T A^T = A^T B^T = I.$$

So  $A^T$  is invert. and in fact.

$$(A^T)^{-1} = (A^{-1})^T.$$

Aside!

$$\underline{Ax \cdot y} = \underline{x \cdot A^T y}.$$

$$\begin{aligned} A & \text{ is } n \times n. \\ x & \in \mathbb{R}^n \\ y & \in \mathbb{R}^n \end{aligned}$$

Ex:  $\begin{pmatrix} 2 & 0 & 3 \\ 0 & 1 & 4 \\ 1 & 3 & 1 \end{pmatrix} \leftarrow \text{this is invertible.}$

row  $\downarrow$

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Conclusion:  $\begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 3 \end{pmatrix}, \begin{pmatrix} 3 \\ 4 \\ 1 \end{pmatrix}$  are

lin. indep. and they span  $\mathbb{R}^3$ .

$\begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 4 \end{pmatrix}, \begin{pmatrix} 1 \\ 3 \\ 1 \end{pmatrix} \leftarrow \text{columns of the transpose so lin. indep.}$

# Invertible Linear Transformations

## Definition

A linear transformation  $T : \mathbb{R}^n \rightarrow \mathbb{R}^n$  is invertible if there is a linear transformation  $S : \mathbb{R}^n \rightarrow \mathbb{R}^n$  such that

$$S(T(x)) = x \text{ and } T(S(x)) = x \text{ for all } x \in \mathbb{R}^n.$$

## Theorem

*A linear transformation  $T : \mathbb{R}^n \rightarrow \mathbb{R}^n$  is invertible iff the standard matrix for  $T$  is invertible.*



$$\begin{aligned} S \circ T &= \text{id} \\ T \circ S &= \text{id} \end{aligned}$$